

CUIWEN ZHU

MC 5474 ♦ f3zhu@uwaterloo.ca

EDUCATION

University of Waterloo 2026 - now
MMath - Combinatorics & Optimization *under co-supervision of Professor Chris Godsil and Professor Jason Bell*
University of Waterloo 2022 - 2026
Bachelor of Mathematics, Honours Pure Mathematics - Combinatorics & Optimization Minor
(Dean's Honours)

EXPERIENCE

Undergraduate Research Assistant, University of Waterloo 2026/05 - 2026/08
MURA funded research with Professor Chris Godsil
Research on the continuous quantum walk, especially on circles with the uniform mixing property.
Part-time Undergraduate Research Assistant, University of Waterloo 2025/10 - 2026/04
part-time research with Professor Chris Godsil
Research on the continuous quantum walk and algebraic graph theory.
Undergraduate Research Assistant, University of Waterloo 2025/05 - 2025/08
MURA funded research with Professor Yu-Ru Liu
Research on number theory, especially on the Hardy–Ramanujan–Littlewood circle method, with application on Waring's Problem, Hasse Principle, and smooth numbers.
Part-time Algebra Tutoring, Tutoring Centre, University of Waterloo 2024/05 - 2026/04
Tutoring first and second year's algebra courses.
Undergraduate Research Assistant, University of Waterloo 2024/01 - 2024/04
MURA funded research with Professor Elana Kalashnikov
Research on algebraic geometry, especially on the coordinate rings of Kronecker quiver moduli spaces.

PROJECTS AND TALKS

“Bypassing the Exact: An Introduction to Quantum Walks and Their Applications” 2026/06/25
Talk at Seminars in Canadian Undergraduate Mathematics Conference in the McMaster University
“The Hasse Principle for Diagonal Forms via the Circle Method” 2025/07/30
Talk at Student Number Theory Seminar in the University of Waterloo
“Circle method on Waring Problem and Hasse Principle” 2025/06/25
Talk at Seminars in Canadian Undergraduate Mathematics Conference in the University of Waterloo
“Quiver Semi-invariants and SAGBI Bases” 2024/07/09
Talk at Seminars in Canadian Undergraduate Mathematics Conference in the University of British Columbia

(CROSSLISTED-)GRADUATE COURSES TAKEN

CO 434: Combinatorial Designs
CO 444: Algebraic Graph Theory
PMATH 441: Algebraic Number Theory
PMATH 445: Representation of Finite Groups
PMATH 446: Introduction to Commutative Algebra
PMATH 450: Measure Theory

PMATH 465: Smooth Manifolds
PMATH 863: Introduction to Lie Groups and Lie Algebras
PMATH 945: Topics in Algebra; topic: Rings and their applications
PMATH 945: Topics in Algebra; topic: Category Theory and Homological Algebra

AWARDS

President’s Research Award	<i>2026</i>
Mathematics Undergraduate Research Award	<i>2026</i>
Mathematics Undergraduate Research Award	<i>2025</i>
President’s Research Award	<i>2024</i>
Mathematics Undergraduate Research Award	<i>2024</i>